The boundary between random and non-random passenger arrivals: Robust empirical evidence and economic implications

Ramandeep Singh*, Daniel J. Graham, Daniel Hörcher, Richard J. Anderson

Transport Strategy Centre, Department of Civil and Environmental Engineering
Imperial College London, United Kingdom

Abstract

In this paper, we investigate the influence of train headways on passenger platform wait times using automated data from the London Underground metro system. For high frequency services, the literature suggests that passenger arrivals are random and that under perfectly random conditions with all other factors held constant, wait times are equivalent to half of the headway between trains. We test this hypothesis using large-scale smart card and vehicle location data, which enables the extraction of access times from total passenger journey times as well as the precise measurement of train headways. Using a semiparametric regression modelling framework, we generate non-linear estimates of the relationship between access times and headway while conditioning for other service supply and demand factors. Marginal platform wait times are then derived numerically via an exposure-response model framework which accounts for potential confounding between the walking and waiting components of access times, thus enabling quantification of the unbiased impact of headways on wait times. For three lines in central London, we observe that marginal wait times transition from greater than half of the headway to approximately one third of the headway as train frequencies decrease. The transition occurs in the range between 2-3 minute headways, lower than earlier estimates in the literature. A series of numerical simulations illustrate the importance of waiting time sensitivity in the optimisation of public transport services. In comparison with the standard wait time assumption, our exercise reveals that the degree of density economies is milder than what the literature suggests, and this may neutralise some of the economic justifications of high public transport subsidies.

Keywords: Platform wait times, AFC data, AVL data, Revealed preference, Semiparametric regression

1. Introduction

The time spent waiting at a station platform prior to boarding is considered by passengers to be one of the most inconvenient phases of a transit journey. In terms of the generalised

*Corresponding author
Email address: ramandeep.singh13@imperial.ac.uk (Ramandeep Singh)

cost of travel, the value of platform wait time is typically weighted at least twice as much as the value of uncrowded in-vehicle time (Wardman et al., 2016). Transport for London applies a higher weighting of 2.5 for wait time, compared to a base weighting of 1 allocated for in-vehicle time in uncrowded conditions (TfL, 2013). Wait time therefore incurs at least double the amount of disutility to passengers as when they are travelling on empty trains. Since wait times are such an influential component of passenger perceptions of transit travel, understanding and quantifying passenger wait time patterns is essential.

Train frequency is a primary determinant of passenger wait time (Newell, 1982; Osana and Newell, 1972). In the literature, the consensus is that passenger arrival patterns are influenced by train frequency levels as follows: (i) when train frequencies are high, passengers arrive in a random manner, resulting in an average wait time equivalent to half of the headway between trains, and (ii) at lower service frequencies, passengers tend to arrive in a non-random manner in order to minimise their wait times. In the most recent empirical work on rail transit covering both high frequency metro and low-frequency suburban rail modes, the transition from random to non-random passenger arrivals is reported to occur in the range between 5 and 11 minute headways (Berggren et al., 2019; Ingvardson et al., 2018; Fan and Machemehl, 2009; Luethi et al., 2007). With increasing access to advanced train arrival information available through open source data online and via mobile phone applications, the transition to timed passenger arrivals may occur at shorter headways.

In this paper, the prevailing hypothesis that passenger wait times transition from random to non-random as headways increase is tested, using the London Underground metro system as a case study. Automated fare collection (AFC) data from the Oyster card system and automated vehicle location (AVL) data on train movements are used, and these data combined provide a revealed preference framework with a very large sample size. Three parts of analysis are undertaken to establish the effect of headways on passenger wait times at the origin station as follows:

1. Total passenger journey times are decomposed via a probabilistic passenger to train assignment algorithm to calculate passenger access time, i.e, the time taken from passenger tap-in to train boarding at the origin station.
2. Semiparametric regression modelling is undertaken to quantify the relationship between headways and access times while conditioning on other service supply and passenger demand characteristics.
3. The effect of headways on marginal passenger wait times is then isolated and quantified via an exposure-response model framework.

Marginal passenger wait times with respect to train headways are calculated in a two-step process. First, expected values of access time at different values of train headway are calculated from the outputs of the semiparametric regression model of access times. This generates a relationship which can be considered analogous to a exposure-response function, a model framework commonly applied in the medical toxicology literature (Haschek et al., 2013; Wang, 2015). In this case, access time is the response and headway is the exposure. Second, the derivative of access time with respect to headway is calculated via

finite differencing. This quantity is used to infer marginal wait time with respect to headway. Through this process, passenger arrival patterns are able to be characterised and the point at which arrivals transition from random to non-random across three high frequency lines in central London is established.

In the literature, the majority of models exploring the relationship between wait times and headway rely on direct measurement of wait times through the observation of samples of boarding passengers at station platforms or from passenger stated preference survey data. Parametric functional forms are assigned to the wait time data, and inferences are then made regarding passenger arrival and wait time patterns (Luethi et al., 2007; Nygaard and Torset, 2016). Separate analyses may then be undertaken to quantify the effect of different supply and demand factors; linear regression modelling and analysis of variance (ANOVA) statistical testing methods are the most prevalent analysis techniques (Fan and Machemehl, 2009; Tavassoli et al., 2018; Berggren et al., 2019).

In comparison with the existing empirical literature on wait times, the methodological contribution of the present paper is threefold. First, we use large-scale automated data sets from which access times are derived in a passenger-to-train assignment framework. This makes data collection more feasible from a time and cost perspective, while the potential bias of small-scale manual sampling is eliminated. Second, the application of semiparametric regression rather than the conventional linear regression methods enables more robust estimation of parameter coefficients which have a greater degree of fidelity to the data. Third, the paper contributes with the derivation of a novel exposure-response functional framework to infer marginal wait times with respect to headway. Within this framework, we quantify the unbiased impact of headways on wait times via the derivation of marginal wait times to account for potential confounding between the walking and waiting components of access times. Our study demonstrates that the numerical approximation of marginal wait times generates valid results for the London Underground metro system, and so the method can be readily applied to other systems to analyse wait time patterns from routinely collected automated data.

Our study also investigates the application of wait time modelling at the supply side of public transport provision. Far-reaching conclusions are derived regarding the economic implications of replacing the standard assumption of random passenger arrivals with robust econometric estimates of the marginal wait time–headway relationship. We show that the simplifying assumption of random arrivals leads to the underestimation of the socially optimal frequency in high-demand periods and overestimation in the off-peak. Moreover, with our new estimates, density economies in the operator costs of service provision tend to disappear, thus neutralising some of the economic justifications for high public transport subsidies.

In terms of the applicability of this study on the demand side of public transport, a number of studies suggest that the provision of real-time information improves passenger perceptions of wait times at transit stops (Dziekan and Kottenhoff, 2007; Watkins et al., 2011; Brakewood et al., 2014a,b). The results of this study reveal the instances where real-time information on train arrivals might be the most useful for passengers. Around the critical headway (approx 2-3 mins), such information is decisive in whether passengers are

willing to risk a non-random arrival. This information could be communicated to passengers, and could ultimately improve passenger perceptions of service quality.

The remainder of the paper is structured as follows: a literature review of work on the estimation of transit platform wait times is presented in Section 2. The study area and data are briefly summarised in Section 3. The methods for the passenger to train assignment algorithm, the semiparametric regression modelling framework, and the estimation of marginal wait times are detailed in Section 4, followed by the results and discussion in Section 5. Conclusions and suggestions for future work are given in Section 6.

2. Literature review

2.1. Fundamental properties of transit wait times

The earliest work on estimating wait times for rail transit was based on work undertaken for bus transit, where researchers made the assumption that passengers arrive randomly, and the resulting wait times follow a uniform distribution (Holroyd and Scraggs, 1966; Osana and Newell, 1972; Abkowitz and Tozzi, 1987).

The seminal formula for platform wait time is:

$$E[w] = 0.5E[h](1 + cv_h^2), \tag{1}$$

where w denotes wait time, h is train headway between consecutive services, and cv_h is the coefficient of variation of headway (Newell, 1982; Osana and Newell, 1972). The formula expects that passengers tend to arrive randomly, and so wait times are a function of half of the headway between trains including an adjustment for the regularity of headways.

O'Flaherty and Mangan (1970), Seddon and Day (1974), Jolliffe and Hutchinson (1975), Turnquist (1978), and Bowman and Turnquist (1981) were among the first researchers to observe that although some passengers do indeed arrive randomly, others use knowledge of schedules via published timetables or experience to adjust their arrival to minimise wait times. In subsequent work on wait times, researchers generally adopt the two-category typology of passenger wait times proposed by Turnquist (1978):

1. Passengers arrive randomly when service frequencies are high and resulting expected wait times are low.
2. Passengers arrive non-randomly and time their arrival to minimise wait time when service frequencies are low and train schedule information is known.

When passenger arrivals are perfectly random with a uniform distribution, the expected wait time $E[w]$ is half of the headway, i.e. $E[w] = 0.5h$. When passenger arrivals are a mix of random and non-random, it follows that the expected wait time can be assumed to be less than half of the headway. Fosgerau (2009) and Fosgerau and Engelson (2011) document the theoretical foundations of the boundary between random and non-random passenger arrivals. In terms of empirical estimates of the transition point, the US Transit Capacity and Service Quality Manual currently quotes the transition at 10 minute headways (Kittelson & Associates Inc et al., 2013). Other empirical research on bus and rail modes

quote values as low as 5 minutes; a brief review of relevant studies are presented in the following sections.

O'Flaherty and Mangan (1970) characterise bus wait times in Leeds, UK, as a linear function of headway, where the expected wait time is defined as $E[w] = 1.79 + 0.14h$. Substituting $E[w] = 0.5$, the distinction between random and non-random arrivals occurs at 5 minute headways. Seddon and Day (1974) analyse bus wait times in Manchester, UK, and also observe a linear function where $E[w] = 2.34 + 0.26h$ and the transition from random to non-random arrivals occurs at approximately 10 minute headways. In more recent work, Nygaard and Torset (2016) study bus travel in Trondheim, Norway and observe non-random arrivals behaviour for bus services with 10, 15, and 20 minute headways. For the 10 minute headway service, the authors specify a linear model for the expected wait time, where $E[w] = 0.32h$. In their study of bus wait times in Austin, Texas, Fan and Machemehl (2009) analyse services with headways ranging from 8 to 60 minutes and also specify a linear model for wait times, where $E[w] = 2 + 0.3h$. The transition from random to non-random arrivals behaviour is reported to occur at 11 minute headways, and it is reported that after a headway of 38 minutes, all arrivals can be considered as non-random.

Two of the most recent studies on rail travel by Luethi et al. (2007) and Ingvardson et al. (2018) use two-part mixture distributions to represent the random and non-random arrivals. Luethi et al. (2007) analyse platform wait times for bus, tram, and rail in Zurich Switzerland, and propose a mixed uniform and Johnson SB distribution. The transition from random to non-random arrivals is observed to occur at approximately 5 minute headways. Ingvardson et al. (2018) study wait times for metro and regional rail lines in Copenhagen with headways ranging from 2 to 60 minutes, and propose a mixed uniform and beta distribution of passenger arrivals. This distribution is tested against the Luethi et al. (2007) mixed uniform and Johnson SB function, and is found to provide a better fit to the wait time data. In terms of the passenger arrivals process, the authors find that the proportion of passengers that exhibit non-random arrival behaviour is estimated at 43% at 5 minute headways, and 52% at 10 minute headways, and so it can be inferred that the transition between random and non-random arrivals occurs between 5 and 10 minute headways. Berggren et al. (2019) observe passenger wait times through the use of a mobile phone application for rail and bus trips in Lund and Malmö, Sweden. The study finds that passengers show evidence of non-random arrival behaviour at 5 minute headways.

2.2. Factors that influence wait times

In line with the seminal formula for wait times for high frequency services widely adopted by operators (refer to equation 1), empirical research verifies that the degree of headway regularity also impacts wait times. Most studies report that waiting times become longer as services become more irregular. Furth and Muller (2006) evaluate waiting costs for short and long headway scenarios while incorporating an allowance for headway reliability, and establish the transition between random and non-random arrivals by equating short headway waiting cost with long headway waiting cost. Using six different operational scenarios, it is shown that waiting costs tend to increase as headways become more irregular. Bowman and Turnquist (1981) further confirm that for observed bus wait times in Chicago, passengers

are more sensitive to the regularity of headways rather than headway itself. Frumin and Zhao (2012) assess wait times on the London Overground rail lines, and find that passengers arrive more randomly when the schedule deviation of the service is high. Nygaard and Torset (2016) assess bus wait times in Trondheim, Norway, and show that passengers experience longer wait times as the deviation between actual and scheduled arrivals increases.

In addition to the influence of headway and headway regularity, empirical work shows that passenger wait times are also influenced by the availability of train timetable information, and other passenger-specific behaviours. Here a brief summary of the some of the recent work in this area is presented.

Ingvardson et al. (2018), Luethi et al. (2007), Frumin and Zhao (2012), and Csikos and Currie (2007) directly and indirectly investigate the impact of passenger information on wait times. Ingvardson et al. (2018) assess rail travel in Copenhagen, Luethi et al. (2007) study bus, tram, and rail modes in Zurich, Frumin and Zhao (2012) study the London Overground rail lines, and Csikos and Currie (2007) investigate rail travel in Melbourne, Australia. The four studies investigate the impact of the time of day and all four report that a higher proportion of passengers arrive randomly in off-peak hours than during the peak. It is argued that those who travel in the peak are more likely to be regular travellers who have knowledge of the train departures, and are therefore more likely to time their arrivals. Ingvardson et al. (2018) also find that a greater proportion of passengers arrive randomly for the metro service, which operates to a set service frequency rather than to a scheduled timetable. In comparison to the random metro arrivals, they observe that passengers time their arrival to minimise wait times for the rail lines where published timetables are available. This illustrates the direct impact of passenger information on wait times; where available, passengers will consult information to time their arrivals in order to reduce wait times.

Passenger-specific behaviours are found to influence wait times in a number of studies. Fan and Machemehl (2009) analyse wait time for buses in Austin, Texas, and perform a linear regression analysis using OLS fitting to investigate the influence of headway, headway regularity, time of day, and individual-specific characteristics including gender, ethnicity, and access mode. The variables found to be significant at 95% are headway, headway regularity, the dummy variable for the off-peak period, and the dummy variable for ethnicity type. All significant variables have positive coefficients and so increases in the value of all variables lead to longer wait times. Berggren et al. (2019) perform linear regression and ANOVA tests on wait times for bus and rail modes in Sweden and observe that individual-specific traits including gender, access mode, trip purpose influence passenger arrival times. Another passenger-specific arrival preference is explored by Leurent and Xie (2018). It is asserted that passengers opt to have longer wait times at the origin station in order to optimally position themselves on the train such that their walking distance at the destination platform is minimised. The method is applied to the Paris metro system using a similar maximum likelihood optimisation algorithm as presented in Xie and Leurent (2017).

Csikos and Currie (2008) investigate passenger-specific behaviours in more depth by observing and categorising passengers according to their arrival preferences. The authors extend on their earlier work on rail wait times in Melbourne, Australia (Csikos and Currie, 2007) and conduct a longitudinal study to analyse the arrival behaviour of regular passengers

who complete the same route for at least 10 weekdays. Four types of arrivals behaviour is defined: 1) those who have the greatest consistency in arrival time and arrive within a narrow window (1-3 minutes before scheduled service), 2) those who arrive randomly but within a consistent wider window of arrival time, 3) those who arrive consistently within a given window, but also display outlying arrival times, and 4) random arrivals. The authors find that passengers with the most consistent and narrow arrival window also exhibit the shortest offset to the train arrival time (i.e. passenger arrival coincides very closely to train arrival), and these passengers also tend to travel earlier in the peak period of the day.

Tavassoli et al. (2018) investigate the impact of a wide range of different factors on passenger wait times for rail travel in Brisbane, Australia. Using a log-logistic accelerated failure time model framework, wait times are assumed to follow a log-logistic distribution and model parameter estimates are assigned a linear form. Model covariates found to be significant include: the time of day (AM peak, PM peak, and other times), operational factors such as headway, the volume of passenger boardings and alightings, the number of platforms, the location of the station, demographic characteristics including population counts and employment, and trip characteristics such as transfer details (if applicable), and if an alternative route is available.

With the exception of Ingvardson et al. (2018), Tavassoli et al. (2018) and Frumin and Zhao (2012), the majority of work on the characterisation of wait time patterns involves collecting wait time data through stated preference surveys or through observation of passengers at stations. Distribution fitting or linear regression methods are then typically adopted to investigate the relationship between wait times and train headways. Studies that assess the impact of additional supply and demand factors aside from headway predominately involve the application of linear regression analysis methods.

In this paper, rather than measuring or obtaining wait time estimates from stated preference sources, the impact of headways on passenger wait times is numerically derived using revealed preference data. A probabilistic passenger to train assignment algorithm is applied to first calculate the access time component of passenger trips from the revealed preference data on total passenger journey times and train movements on the London Underground. The non-linear semiparametric regression method is then applied to estimate access times conditional on headways and other supply and demand characteristics of the system. Marginal wait times as a function of headway are then numerically derived, enabling an analysis of passenger wait time patterns at different train service frequencies.

2.3. Applications of wait time modelling

In various fields of transport science, the standard assumption that wait time is equivalent to half of the headway is typically applied, and so replacing this standard assumption has a number of implications. Demand prediction and policy appraisal is among the most straightforward applications. Interventions that affect the frequency of service provision have an impact on waiting times as well, and empirical evidence shows that public transport demand is highly sensitive to this journey attribute (Paulley et al., 2006). The unit value of waiting time is significantly higher than the value of standard in-vehicle travel time; the corresponding multipliers reach 1.4–1.8 in the literature (Wardman, 2004). If the impact

of increased frequency on demand is significantly higher (lower) than the standard wait time assumption predicts, then the corresponding response on the demand side might be substantially underestimated (overestimated) by demand models. This implies that the benefit side of such interventions is also underestimated (overestimated) in policy appraisal, thus leading to biased recommendations in project ranking.

A similarly influential application of wait time models is in short-run supply optimisation. Waiting time is one of the key cost components responding to the frequency of service. The trade-off between the operator costs on the one hand and the waiting time savings on the benefit side is in the heart of supply modelling since the seminal contribution of Mohring (1972). The *marginal* waiting time saving has an especially relevant role here, as even the most recent literature still relies on the assumption that it equals to one half (for example, Jara-Díaz and Gschwender (2003); Hörcher and Graham (2018)).

Let us consider the simple example of frequency optimisation in the original model of Mohring (1972). The social cost of service provision is the sum of waiting time and operating costs:

$$C(f) = Q \cdot \gamma w(f) + o(f), \tag{2}$$

where Q is the number of users, γ is the value of waiting time, and $o(f)$ is a frequency-dependent operating cost. Thus, the first-order condition of cost minimising optimal frequency provision is

$$o'(f) = -\gamma Q \cdot w'(f) = Q \cdot \frac{\partial w}{\partial h} f^{-2}. \tag{3}$$

That is, the marginal operator cost equals the marginal waiting time benefit with respect to frequency. Now, let us consider the relationship between waiting time and headway, and the fact that $h = f^{-1}$. This implies that

$$w'(f) = \frac{\partial w}{\partial h}\frac{\partial h}{\partial f} = -\frac{\partial w}{\partial h} f^{-2}. \tag{4}$$

This amends the first-order frequency condition in equation (3) to

$$o'(f) = \gamma Q \cdot \frac{\partial w}{\partial h} f^{-2}, \tag{5}$$

and, after rearrangement, the optimal frequency becomes

$$f = \sqrt{\frac{\partial w}{\partial h}\frac{\gamma Q}{o'(f)}}. \tag{6}$$

Mohring as well as subsequent contributions in the literature assume $w = 0.5h$, so that $\partial w/\partial h$ in the frequency formula is a constant, 0.5. However, if the assumption of random arrivals does not hold any more, and the marginal waiting time benefit is a function of frequency itself, then equation (6) above, often mentioned as the *square root principle*,

becomes an implicit function of frequency. With the false assumption of random arrivals, this analytical result suggests that the optimal frequency is downward biased if $\partial w/\partial h$ is in fact higher than 0.5, and upward biased if $\partial w/\partial h < 0.5$.

In section 5.3 we investigate the impact of the present paper's wait time estimates on optimal frequency provision and subsidisation.

3. Study area and data

Selected sections of the Central, Jubilee, and Victoria lines on the London Underground are analysed as follows: the entire length of the Victoria line (16 stations); West Acton to Oxford Circus on the Central line (12 stations); and Bond Street to North Greenwich on the Jubilee line (10 stations). The experimental area is depicted in Figure 1. We select origin-destination pairs on each line that represent single-line trips only, excluding transfers, and the line sections are chosen such that route choice decisions do not need to be made under regular operating conditions. Since the line sections cover some of the busiest routes in central London, the passenger data capture a mix of regular work commuters, school children, tourists, and others.

Passenger trip data and train movement data over a period of 7 weeks from October to December in 2013 are used, focusing on weekday travel (Monday to Friday) only. The passenger trip data are recorded via the Oyster smart card system, and include the timestamps and locations of each trip between the origin and destination stations. The train movement data are recorded by the internal TfL NetMIS system and include departure timestamps and locations of each train movement at each station platform. The train movement data are used to calculate the headways associated with each train trip; headways are calculated as the difference between departure timestamps of consecutive train services at a given station platform.

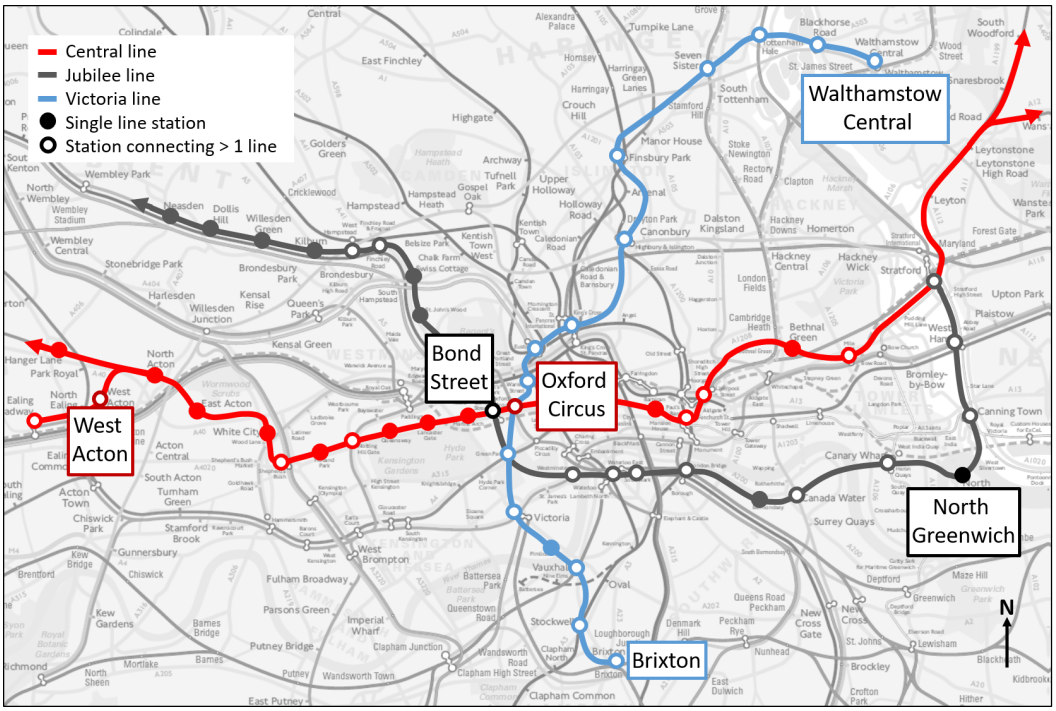


Figure 1: Study area, London Underground (adapted from TfL, 2014)

4. Methods

The analysis involves three main parts. First, journey times from the AFC data set report the total journey times only, and so to enable the journey times to be split into parts, passengers must be allocated to trains. This is achieved by merging the AFC trip data with the AVL train movement data and applying a probabilistic train assignment algorithm, as presented in section 4.1. Second, a semiparametric regression model is developed to quantify the relationship between access times and headway while conditioning for other potentially influential supply and demand factors, as detailed in section 4.2. Third, the unbiased relationship between access times and headway as previously quantified is used to generate an average exposure-response model of access times, with headway as the sole exposure variable. Two exposure-response models are generated: (i) environmental which represents the relationship between access times and headway including the effect of all other supply/demand covariates, and (ii) base which represents the relationship between access time and headway only. The exposure-response models are then used to compute the derivative of access times with respect to headway, and this quantity represents marginal platform wait times. Environmental and base models of marginal platform wait times with respect to headway are derived, as presented in section 4.3. For clarity, a summary of the modelling framework including references to relevant equations is illustrated in Figure 2

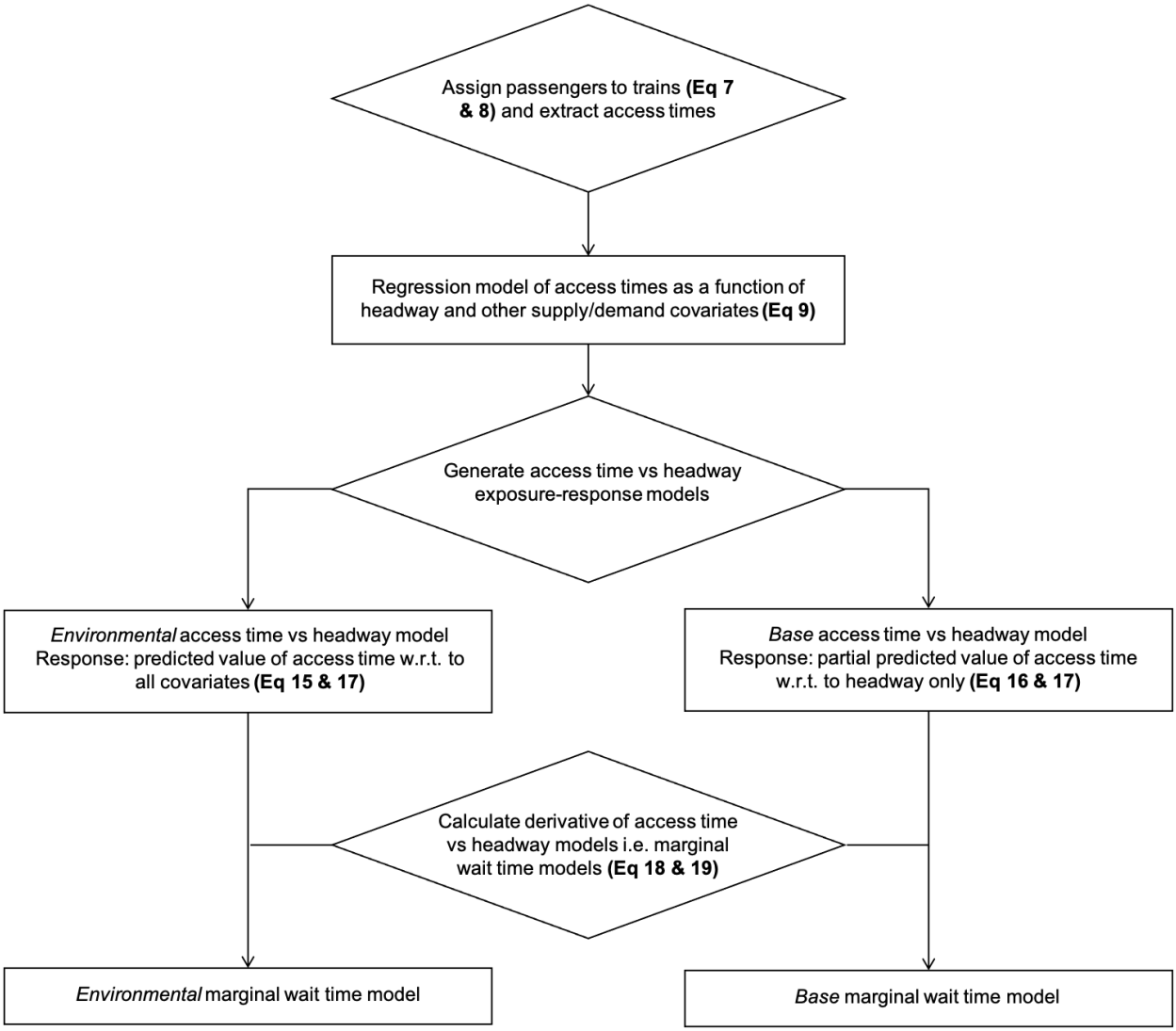


Figure 2: Outline of modelling framework

4.1. Passenger to train assignment

Full details of the assignment method applied in this analysis are available in Singh (2019). The following section provides a summary of the algorithm.

Let t_i^{entry} and t_i^{exit} denote the entry and exit times of passenger i as recorded by the passenger trip AFC data. DT_{oj} is the departure time of train j at the origin station o, and DT_{dj} is the departure time of train j at the destination station d as recorded by the train movement AVL data. For train j to be considered as a *feasible* itinerary for passenger i:

1. Passenger i must enter the system before train j departs at the origin: $t_i^{entry} \leq DT_{oj}$.
2. Passenger i must exit the system after train j departs at the destination: $t_i^{exit} \geq DT_{dj}$.

The total number of feasible train itineraries is n_j, where $j = 1, .., n_j$ denote the feasible train itineraries. By applying the passenger entry and exit conditions, the passenger trips are categorised as follows:

1. *Unambiguous* trips: Train j is the only feasible train for passenger i, i.e. $t_i^{entry} \leq DT_{oj}$ & $t_i^{exit} \geq DT_{dj},\ n_j = 1$. The set of unambiguous trips is referred to as $\mathcal{U}$.
2. *Ambiguous* trips: Passenger i can feasibly board more than one train, i.e. $t_i^{entry} \leq DT_{oj}$ & $t_i^{exit} \geq DT_{dj},\ n_j > 1$. The set of ambiguous trips is referred to as $\mathcal{A}$.

The set of ambiguous trips $\mathcal{A}$ are allocated by calculating the Bayesian probabilities of the egress times associated with each feasible train itinerary y_{ij}^{eg} as follows:

$$P(y_{ij}^{eg}|r_k^{eg}) = \frac{P(r_k^{eg}|y_{ij}^{eg})P(y_{ij}^{eg})}{P(r_k^{eg})}, \qquad i,j \in \mathcal{A}, k \in \mathcal{U}_{op,od} \tag{7}$$

where y_{ij}^{eg} is the egress time for passenger i and train itinerary $j = 1,..n_j$ in the set of ambiguous trips, i.e. $i,j \in \mathcal{A}$, r_k^{eg} is a randomly sampled egress time from the set of unambiguous trips in the off-peak period for the given OD pair, i.e. $k \in \mathcal{U}_{op,od}$ where $\mathcal{U}_{op,od} = \mathcal{U}_{od}(t_i^{entry} < 7\text{am or } 10\text{am} < t_i^{entry} < 4\text{pm or } t_i^{entry} > 7\text{pm})$. $P(y_{ij}^{eg})$ is the probability of observing the egress time for the itinerary of interest, as generated via the kernel density probability function of the egress times of unambiguous trips, $P(r_k^{eg})$ is the probability of observing the randomly sampled egress time, as generated via the kernel density probability function of the egress times of unambiguous trips, and $P(r_k^{eg}|y_{ij}^{eg})$ is the probability of observing the randomly sampled egress time, as generated via the kernel density probability function of the egress times of unambiguous trips in the off-peak period.

The passenger is then assigned to the train itinerary with the highest probable egress time as per equation (8):

$$P(y_{ij^*}^{eg}) = \max_{1 \leq j \leq n_j} (P(y_{ij}^{eg}|r_k^{eg})), \qquad i,j \in \mathcal{A}, k \in \mathcal{U}_{op,od}, \tag{8}$$

where $y_{ij^*}^{eg}$ is the egress time of the passenger for the assigned to train j^*.

4.2. Access time regression model

Semiparametric regression is undertaken to quantify the relationship between access times and headway. To ensure that the headway covariate is not acting as a proxy for other factors, we also condition for a number of potentially influential supply and demand characteristics. This model then enables the isolation of the sole impact of headways on access times, which is then subsequently used to quantify the impact of headways on arrival patterns.

The advantage of the semiparametric regression method is that the relationships between the independent and dependent variables are defined via basis functions generated from the data, therefore enabling non-linearities and potential discontinuities to be modelled (Wood, 2006). The basis functions are penalised thin plate regression splines, which are interpolation functions that are assigned to match the observed data points with a penalty applied to capture the trade-off between the degree to which the spline functions match the data and the degree of smoothness. The modelling is undertaken using the R statistical software package 'mgcv'. The models are fitted using penalised iteratively reweighted least squares

(PIRLS) and the restricted maximum likelihood (REML) technique is used to estimate the model parameters (Wood et al., 2015; Wood, 2006).

For each passenger trip i assigned to train itinerary j, an access time regression model can be formulated as a function of headway h_{ij} and other service supply and demand covariates V_{ij}, and estimated model parameters $\widehat{\theta}$. The headway and other service supply and demand covariates are modelled by applying the following forms: continuous variables are modelled with either a parametric or non-parametric form, and categorical factors are modelled with group-specific fixed effects. The general form of the access time regression model is as follows:

$$\begin{aligned} y_{ij}^{ac} &= m(h_{ij}, V_{ij}, \widehat{\theta}) + \epsilon_{ij} \\ &= \alpha + X_{ij}^T \beta + \sum_{k=1}^{K} f_k(X_{ij}^*) + Z_{ij}^T u + \epsilon_{ij} \end{aligned} \tag{9}$$

where y_{ij}^{ac} is the access time, α is the model constant, β are the parameter coefficient estimates for the covariates X_{ij} modelled parametrically, $f_k, k = 1..K$ are the smooth basis functions based on penalised thin plate regression splines of the covariates X_{ij}^* modelled non-parametrically, u is a vector of estimated group-specific fixed effects for the categorical factors Z_{ij}, and ϵ_{ij} is the random error term, assumed to be independently and identically distributed with mean 0 and given variance σ_ϵ^2, such that $\epsilon_{ij} \sim \mathcal{N}(0, \sigma_\epsilon^2)$.

The data set for the regression model consists of all trips within the defined study area. The resulting error term therefore captures the residual variance across all trips, after conditioning for the covariates and fixed effects in the model. The dependent and independent variables are defined further in the following sections.

Access time is the dependent variable y_{ij}^{ac} in equation 9 and is defined as the time from passenger tap-in at the origin station to the departure of the assigned train from the origin station platform, as per equation 10. It should be noted that access time is measured in minutes.

$$y_{ij}^{ac} = DT_{oj} - t_i^{entry} \tag{10}$$

The independent variables in the model can be categorised into those relating to supply characteristics and those relating to demand characteristics, and are discussed in more detail in the following sections. In terms of notation mapping to equation 9, the parametric and non-parametric continuous covariates denoted by X_{ij} and X_{ij}^*, respectively, are headway, the coefficient of variation (COV) of headway, headway normalised by mean headway, passenger volumes, passenger trip frequency, and the paired time component. It should be noted that the continuous covariates are all initially modelled with non-parametric forms, and are subsequently assigned parametric forms if appropriate; this is discussed further in the modelling results. The categorical variables denoted by Z_{ij} are time of day, station and line effects, and passenger card based categories.

4.2.1. Supply side covariates

Headway - To accommodate differing demand levels, the headways between trains vary across times of the day and across locations. As discussed in the literature review, headway is the primary determinant of platform wait times, and in turn, passenger access times. In the model, the headway for each trip corresponds to the train the passenger boarded as assigned via the passenger to train assignment algorithm, and is measured in minutes.

Coefficient of variation of headway - In line with the seminal formula for platform wait times presented in the literature review, the COV of headway is included in the access time model to capture headway regularity. The COV of headway is calculated over a 15 minute period for each OD pair.

Headway normalised by mean headway - This covariate is included to capture potential effects of train bunching. Train bunching refers to the condition where headways become shorter as a result of longer train dwell times at stations. Bunching is more likely to occur under congested conditions, where longer dwell times are required for higher volumes of passenger boardings and/or alightings. The value is calculated as the headway associated with the given trip normalised by the mean value of headways over the corresponding 15 minute period for each OD pair.

Time of day - This covariate captures any unobserved temporal variation in the model. A total of 72 categories are modelled to represent four 15 minute intervals per hour over the operating period from 6am to 12am. A fixed effects form is assigned to accommodate potential correlation with the other dynamic covariates in the model.

Station and line effects - A set of static covariates capture any unobserved group-specific effects of lines and stations. There are 35 stations and 6 combinations of the three lines in two directions of travel. These effects are modelled as fixed effects, thus accommodating potential correlation with the other station and line-based covariates in the models.

4.2.2. Demand covariates

Passenger volumes - Passenger density indicators are included to capture the effects of passenger congestion on access times. It is estimated that trips made using the Oyster card represented 70% of all trips in 2013 with the reminder undertaken using alternate ticketing modes, such as paper tickets and travel cards (Uniman, 2009; Paul, 2010). Therefore to represent total passenger volumes including non-Oyster trips, the TfL Rolling Origin and Destination survey (RODs) database is used to calculate the passenger density indicators.

Two passenger volume indicators are included in the regression model: platform loading at the origin station, and inter-station line loading. The general equation is given in 11, where d_{qt} denotes the passenger density indicator for the location of interest q i.e. platform loading, or line loading, n_{qt} is the number of trips during time period t, normalised by the average number of trips $N_{qt,day}$ per 15 minute period t over the day.

$$d_{qt} = \frac{n_{qt}}{N_{qt,day}} \tag{11}$$

Passenger trip frequency - As discussed in the literature review, route familiarity can have an effect on walking and waiting times, with regular commuters who are more familiar

with their route typically having shorter journey times. To capture this potential effect, the number of repeat trips a passenger makes on a given OD route is included in the models.

Passenger card-based categories - The Oyster card records for each trip include information on passenger age, ticket type, and discount type. As mentioned in the literature review, individual-specific characteristics may have an effect on journey times, and so the aforementioned characteristics are included in the access time model. There are 4 age categories, 13 ticket types, and 4 discount types. The passenger categories are modelled as fixed effects to accommodate potential correlation between the multiple covariates representing passenger characteristics.

Paired time component - In the literature it is hypothesised that passengers may deliberately position themselves at the origin platform such that they reduce their egress distance and walk time to the exit of the platform at the destination station (Leurent and Xie, 2018). To account for the potential dependency between access and egress times, the paired egress time component for each trip is included in model.

The selection of the fixed effects structure for time, station, line, and passenger attributes is further verified in the discussion of the access time model results in section 5.1.

4.3. Derivation of marginal platform wait times

As defined in the previous section, access time is the time taken from passenger tap-in at the origin station to the point where the train departs from the origin station platform. There are two distinct phases of passenger movements during this time: (i) the time taken to walk from the ticket gates to the platform, z_i and (ii) the wait time at the platform before boarding, w_{ij}. Based on this, access time can therefore be expressed as per equation (12).

$$y_{ij}^{ac} = DT_{oj} - t_i^{entry} = w_{ij} + z_i \tag{12}$$

Since we are interested in interpreting the impact of headways on wait times, we can further break down the wait time and walk time components to consist of a base component and an adjustment for differing environmental conditions arising from changes in headway. The base component of wait times (w_{ij}^b) can be considered as solely being a function of headway, while the environmental component (w_{ij}^e) captures wait time effects that may arise at the platform due to headway regularity, passenger volumes, and individual passenger behaviours. It would be reasonable to assume that the environmental component of wait time is a function of headway, for example, we expect higher degrees of passenger loading and headway irregularity during peak times when headways are shorter. The walk time component can be similarly broken down to consist of the base time for a passenger to walk from the entry gates to the platform in uncongested conditions (z_i^b), plus as an adjustment to account for differing environmental conditions due to passenger volumes and individual passenger behaviours en route to the platform (z_i^e). The base component of walk time can be considered independent of headway, since this component is solely dependent on the fixed distance between the entry gates and platform. As for the environmental component of wait time, it would be reasonable to assume that it is a function of headway.

For calculation purposes, equation (12) can be expressed in terms of the expectations of the variables, and the wait and walk time components can be further expanded into base and environmental components as described:

$$E[y_{ij}^{ac}] = E[w_{ij}] + E[z_i] = E[w_{ij}^{b}] + E[w_{ij}^{e}] + E[z_i^{b}] + E[z_i^{e}] \tag{13}$$

Considering the sole influence of headways h_{ij} on access times, the derivative of access time with respect to headway is expressed as:

$$\frac{\partial E[y_{ij}^{ac}]}{\partial h_{ij}} = \frac{\partial E[w_{ij}^{b}]}{\partial h_{ij}} + \frac{\partial E[w_{ij}^{e}]}{\partial h_{ij}} + \frac{\partial E[z_i^{b}]}{\partial h_{ij}} + \frac{\partial E[z_i^{e}]}{\partial h_{ij}} = \frac{\partial E[w_{ij}]}{\partial h_{ij}} + \frac{\partial E[z_i^{e}]}{\partial h_{ij}} \tag{14}$$

Since the base component of passenger walk time is independent of the headway between trains, the rate of change in base passenger walk time with respect to headway $\frac{\partial E[z_i^{b}]}{\partial h_{ij}}$ in equation (14) is omitted. The derivative of access time with respect to headways therefore primarily represents the marginal wait time at the platform $\frac{\partial E[w_{ij}]}{\partial h_{ij}} = \frac{\partial E[w_{ij}^{b}]}{\partial h_{ij}} + \frac{\partial E[w_{ij}^{e}]}{\partial h_{ij}}$ along with the environmental component of walk time $\frac{\partial E[z_i^{e}]}{\partial h_{ij}}$. In the analysis, our primary interest is to quantify marginal wait time with respect to headway. However, from the regression model of access times, it is difficult to isolate the marginal wait time component from the marginal environmental walk time component. As such, we instead present two functions for marginal wait time from the predicted values of the access time regression model: (i) environmental marginal wait times, and (ii) base marginal wait times.

The environmental model is generated by using the predicted values of access times at the scale of the response with respect to all covariates. In this case, the predicted values of access time are extracted as follows:

$$\widehat{y}_{ij}^{ac} = m(h_{ij}, V_{ij}, \widehat{\theta}), \tag{15}$$

where $\widehat{y}_{ij}^{ac}$ is the predicted value of the access time regression function with respect to all covariates i.e. headway, and the supply and demand covariates denoted collectively by V_{ij}, and model parameter estimates $\widehat{\theta}$. The derivative of this quantity with respect to headway is equivalent to the marginal wait time and marginal environmental walk time as specified in equation 14, and therefore includes the impact of environmental conditions such as differing levels of congestion and individual-specific behaviours, which can result in faster or slower access times.

The base model is generated by extracting the partial predicted values of access times at the scale of the linear predictor with respect to the headway covariate only, i.e.:

$$\widehat{y}_{ij}^{ach} = m(h_{ij}, \widehat{\theta_h}), \tag{16}$$

in which $\widehat{y}_{ij}^{ach}$ is the partial predicted value of the access time regression function with respect to headway and the model parameter estimates associated with the headway covariate are denoted by $\widehat{\theta_h}$. The derivative of this quantity with respect to headway is the base component of marginal wait time $\frac{\partial E[w_{ij}^{b}]}{\partial h_{ij}}$, which represents the sole impact of headway only.

Finite differences between each pair of sequential data points are then required to be taken in order to compute the derivative of access times with respect to headways. Due to the large number of observations and high levels of dispersion in the predicted values generated from the access time model, the finite differences between a set of two consecutive individual points contain extreme outlying values which tend to artificially skew the results. In order to smooth out the outliers, the mean of the predicted values is calculated for a fixed value of headway at 1 second intervals. The 1 second interval corresponds to the equivalent level of accuracy in the train movement data timestamps. Repeating this calculation for the range of headways observed allows the generation of an exposure-response function, where headway is the exposure variable and the expected value of access time is the response.

Two steps are involved to compute the derivative of access times with respect to headways. The first step is the calculation of the exposure-response function of expected access times with respect to headway, which is given in general terms by

$$E[y^{ac}(h)] = \frac{1}{n} \sum_{h_c=h}^{h+\Delta} \sum_{ij=1}^{n} \widehat{p}_{ij}^{ac}(h_c), \tag{17}$$

where $E[y^{ac}(h)]$ is the expected value of access times with respect to headway h in minutes. The term $\widehat{p}_{ij}^{ac}(h_c)$ represents predicted values of access times at a given point value of headway h_c. The summations refer to the summation of access times across all n number of passenger and train itinerary combinations ij over a 1 second interval (Δ) of headways. The bounds for headway are defined from $h = 1$ to $h = 9$ minutes 59 seconds, which represents approximately 99% of observations in the raw data set after excluding upper and lower outlying values of headway.

The second step is the estimation of the derivative of access times with respect to headways. The derivative is estimated through a discrete approximation by computing finite differences between the average predicted values calculated in equation 17. This process is undertaken for the total data set, and separately for peak and off-peak periods. Four models are estimated, segmented by time period: (1) all time periods, (2) peak of the AM peak, (3) inter-peak, and (4) peak of the PM peak. That is,

$$\frac{\partial E[y^{ac}(h)]}{\partial h} = \frac{E[y^{ac}(h+a)] - E[y^{ac}(h)]}{(h+a) - h}, \tag{18}$$

where $E[y^{ac}(h)]$ is the expected value of access time with respect to headway h according to equation 17, and a is a small finite increment.

5. Results

5.1. Access time regression model

The following sections present the results of the access time regression model. First, the data cleaning process is summarised, second, the modelling process to reach the final model form is presented, and third, the results of the significant covariate elasticities are discussed.

5.1.1. Data

The raw data set including all outliers was initially modelled, and for a number of covariates, outlying values were observed at the extremities. The outliers were therefore discarded, and the resulting ranges of the data set are summarised in Table A.7 in Appendix A.

Due to the numerous covariates in the model, the filtered data set is tested for the presence of linear and monotonic correlations through the Pearson and Spearman correlation tests, respectively. The maximum values of the correlation coefficients are observed between headway and normalised headway. The Spearman correlation coefficient has a maximum value of 0.76 and the Pearson correlation coefficient has a maximum value of 0.71. In correlation testing, the interpretation of the degree of association between two variables is context dependent; a correlation coefficient of 0.85-0.90 would indicate a strong correlation in cases where there are many data points (Kutner et al., 2005; Aldrich, 1995). In this analysis, there are approximately 5 million data points, and so we can infer that the correlations are not strong. It is therefore appropriate to include all covariates in the model. Moreover, we highlight that potential collinearity issues would be detected in the regression modelling, but no such issues are observed in the final model output.

5.1.2. Model form

As previously presented in the regression model framework, the continuous covariates can be modelled as either non-parametric smooth terms or as parametric terms, and the categorical factors are modelled with group-specific effects. All continuous variables are initially modelled with non-parametric smooths, and if the functions are found to approximate known parametric forms, a parametric form is assigned for tractability and model parsimony. For the categorical factors capturing group effects, either a fixed or random effects structure can be assigned. The fixed effects structure allows for the group effects to be correlated with the other covariates in the model, while the random effects do not allow for correlation with other covariates (Searle et al., 1992). From a theoretical standpoint, a fixed effects structure is most appropriate for the group effects as there are many covariates concurrently capturing similar characteristics of the system, and this could result in a degree of correlation between covariates. For verification purposes, however, a random effects structure is also trialled.

The results of five model forms are presented in Table A.8 in Appendix A to demonstrate the model selection process. Model 1 represents a basic reference model with access time as the response and headway as the sole covariate, which is modelled with non-parametric smooths. The remaining models trialled include all additional supply/demand covariates as follows: Model 2 - All continuous covariates modelled with non-parametric smooths and fixed group effects, Model 3 - All continuous covariates modelled with non-parametric smooths and random group effects, Model 4 - All continuous covariates modelled with linear relationships and fixed group effects, and Model 5 - Final model form with parametric and smooth continuous covariates and fixed group effects. To evaluate the performance of the models, four criteria are used (Wood, 2006): the adjusted coefficient of determination ($R^2_{adj.}$), the percentage of explained model deviation ($D_{explained}$), the Akaike Information Criterion (AIC), and the restricted maximum likelihood (REML) score. It should be noted

that the Hausman test is conventionally used to compare the fixed and random effects structures for simple linear panel models, however, this test is not able to be applied to the semiparametric models generated here. As such, the performance of the fixed and random effects are evaluated through the goodness-of-fit indicators.

According to the goodness-of-fit results, all models with additional supply/demand covariates outperform the reference model with headway only, as expected, thus indicating the benefit of including additional supply/demand covariates alongside headway. The difference between the fixed and random effects structures (models 2 and 3) is not perceptible, and so in keeping with the theoretical justification, a fixed effects structure is used. In models 2 to 5, for the form of the continuous covariates, the goodness-of-fit of the fully non-parametric model form 2 is best while the linear model form 4 performs worst. In developing the final model form, a balance between maximising the goodness-of-fit while maintaining model parsimony and tractability is desired. As such, the final model form 5 performs better than the simple linear model form 4, and in line with the fully non-parametric model form 2.

To verify that the relationship between access times and headway is robust to model form, we illustrate the relationship for each trialled model form 1 to 5 in Figure A.10 in Appendix A. In Table A.8, headway is significant at a level of $\alpha = 0.001$ in all models. As shown in Figure A.10, all models illustrate an approximately monotonically increasing relationship between access time and headway. Models 2, 3, and 5 (final form), which include additional supply/demand covariates, all generate equivalent forms, thus indicating the robustness of the access time and headway relationship when other covariates are assigned different forms. As discussed previously, the linear model form (model 4) oversimplifies the relationship; Figure A.10 shows that it does not capture the slight flattening of the slope at higher values of headway. In model 1 (headway as the sole covariate), there is a plateau at headway values of 1.6-2.0 minutes, while models 2, 3, and 5 do not have the same form. The highest scheduled operating frequency of trains within the study area is 35 trains per hour on the Central line, and this corresponds to a minimum headway of approximately 1.67 minutes between trains. Approximately 15% of all trips in the data set are undertaken when trains operate at headways less than 1.67 minutes, and so the plateau in model 1 is most likely representing the transition from bunched train operations in highly congested conditions at headways shorter than the published minimum, to regular train operations in line with the published schedules. Considering this, it is likely that the addition of covariates representing headway regularity and passenger volumes in models 2, 3, and 5 has addressed potential confounding with headway, and corrected the plateau observed in model 1.

For the final model form 5, the explanatory power of the model is relatively low as indicated by the $R^2_{adj.}$ and $D_{explained}$ values of 0.16. Though we have included 10 additional covariates that are most likely to have a confounding effect on the relationship between access times and headways, there may be additional missing covariates that explain the additional variation in waiting and walking times. It should be noted that though the explanatory power of the model is relatively low, this does not impact the primary motive of the model, which is to quantify the relationship between access times and headways. As mentioned, the headway covariate is significant at a level of $\alpha = 0.001$, furthermore, we have included 10 additional supply/demand covariates that are most likely to have a confounding effect

on headway, and we have demonstrated that the relationship between headway and access time is robust to the model forms trialled. Therefore, we can infer that the model results are robust for further analysis.

It should be noted that another reason for the relatively low explanatory power may be a consequence of the Oyster card timestamp precision. The Oyster timestamps are accurate to one minute only. As information on the magnitude of the inaccuracy for each timestamp is unknown, the inaccuracy can be assumed to follow a uniform random distribution between 0 and 59 seconds, and this is captured by the random error term of the access time regression model. Furthermore, any random error in the passenger to train assignment algorithm would also be captured in the random error term of the model. It is worth highlighting that potential impacts on the random error term do not impact the magnitude of the parameter estimates, and so the relationship between access times and headways would remain robust. We acknowledge that if there are systematic errors in the timestamp precision and assignment, this would result in biased estimates.

5.1.3. Covariate significance and elasticities

The results presented in this section refer to the final model form (model form 5, Table A.8) for the access time regression model as discussed in the previous section. For the covariates measured on a continuous scale, the form, significance, and mean elasticity values are given in Table 1. For the fixed effects terms, the significance and maximum and minimum values of the fixed effects are given in Table 2. Due to the large number of observations in the model (approximately 5 million points), a minimum significance level of $\alpha = 0.01$ (99%) is adopted.

All terms except all passenger card types, some categories of passenger discount type, and some time intervals are insignificant at a level of $\alpha = 0.01$ (99%). For the passenger card types and the insignificant categories of passenger discount type, it is likely that passenger ages and significant discount types capture these effects. The insignificant time intervals are likely captured by the other time-varying covariates in the model.

Table 1: Results for continuous model covariates

Covariate	Form	Sig.	Mean elasticity
Service capacity covariates - dynamic			
Headway	splines	***	0.37
COV headway	poly deg 3	***	0.06
Normalised headway	splines	***	-0.19
Demand covariates - demand volumes			
Origin platform loading	poly deg 2	***	0.05
Line loading	poly deg 3	***	0.09
Demand covariates - passenger characteristics			
Passenger trips	splines	***	-0.07
Egress time	poly deg 2	***	-0.20
Mean access time (mins)	3.49		

Sig. notation: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
Note: Covariate elasticities are unit-less

Table 2: Results for fixed effects

Fixed effect	Sig.	Min. value	Max. value
Service characteristics			
Time	Variable	-0.62	0.43
Station	***	-2.22	0.77
Line/direction	***	-0.66	1.15
Demand - passenger characteristics			
Passenger age	***	0.00	1.24
Passenger discount	Variable	0.00	0.07
Passenger type		-	-
Model intercept	***	4.69	
Mean access time		3.49	

Sig. notation: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
Note: Fixed effects are measured in minutes

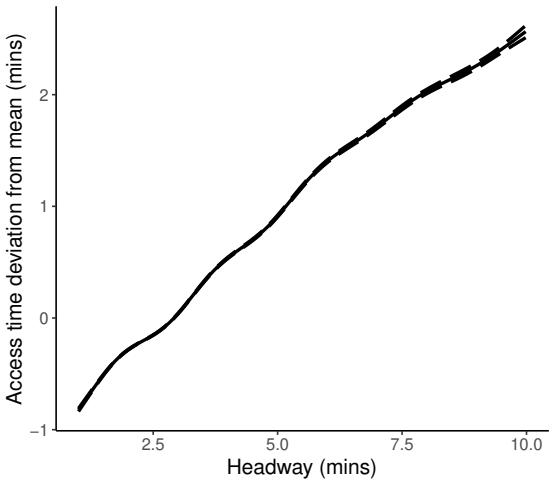


Figure 3: Access time with respect to headway

The purpose of the analysis is to investigate the relationship between headway and platform wait times, and so headway is the primary covariate of interest in the access time regression model. As shown in Table 1, headway is significant at a level of $\alpha = 0.001$. Furthermore, out of all significant dynamic covariates in the access time model, headway is the most influential, with an average elasticity of 0.37. The positive elasticity implies that as headways increase, access times also tend to increase, and Figure 3 shows that access times tend to increase monotonically with headway.

In terms of the covariates that capture the regularity of headways, both the COV of headway and the measure of headway normalised by mean headways are significant in the model. This aligns with the findings in the literature; the COV of headway is included in the seminal formula for passenger wait times (Newell, 1982; Osana and Newell, 1972) and the degree of consistency in headways is further found to influence wait times in empirical studies by Frumin and Zhao (2012) and Nygaard and Torset (2016). The COV of headway is monotonically positive, suggesting that as COV increases, access times also increase with an average elasticity value of 0.06. The covariate capturing normalised headways has a monotonically negative elasticity, with an average value of -0.19. The indicator represents the degree of train bunching, and can be interpreted as capturing the degree of congestion in the system; low values of the indicator (< 1) imply congested train operations, while high values (> 1) imply less congested conditions. Therefore, the negative elasticity plausibly indicates that as train congestion alleviates, access times tend to decrease.

The covariates that represent passenger demand indicate congestion levels at the platform and line loading. Both covariates have monotonically increasing elasticity values, which plausibly suggests that access times tend to increase as passenger volumes increase. In terms of the average elasticity values, the line loading indicator has a value of 0.09 and the platform indicator has a value of 0.05, which suggests that line loading is more influential on access times.

The passenger trip frequency covariate illustrates the impact of route familiarity. The elasticity monotonically decreases as trip frequency increases with an average elasticity of

-0.07. This suggests that passengers who complete more repeat trips on a given route are likely to have faster access times. Another passenger-specific behavioural characteristic is the paired egress time for each trip. As mentioned previously, there may be a propensity for passengers to minimise their egress times at the destination station by trading off their access time at the origin station, thereby inducing some dependence between the two components (Leurent and Xie, 2018). The results show that there is evidence of an inverse interaction between the components; access times decrease monotonically as egress times increase with an average elasticity of -0.20.

In terms of the group-specific fixed effects in the model, the fixed effects for stations range from -2.22 to 0.77 minutes, and the line/direction effects range from -0.66 to 1.15 minutes (refer to Table 2). The static characteristics of individual stations therefore tend to be more influential on access times, rather than the static characteristics specific to lines and directions. The effects of the stations seem to reflect the complexity of station layouts. For example, the station with the lowest effect (thus quickest access times) is White City, which has one of the shortest average walking paths to the platform, while Euston has one of the longest average walking paths to the platform and also has the highest effect (thus slowest access times).

The fixed effects for time are also significant, indicating that there are residual time effects in addition to the time-varying impacts captured by the other dynamic covariates in the model. Figure 4 illustrates the magnitude of the fixed effects in each 15 minute time interval. The majority of the effects are negative, resulting in a reduction in access times. The fixed effects for only two intervals in the AM peak between 8:15 and 8:45am are positive and result in longer access times. The time effects appear to capture demand and operational patterns; the effects tend to increase in the AM and PM peak periods and at the extremities of the day, and tend to be lower during the midday and evening off-peak periods. The time periods between 5:45pm and 6:30pm are not significant at a level of 99% in the model.

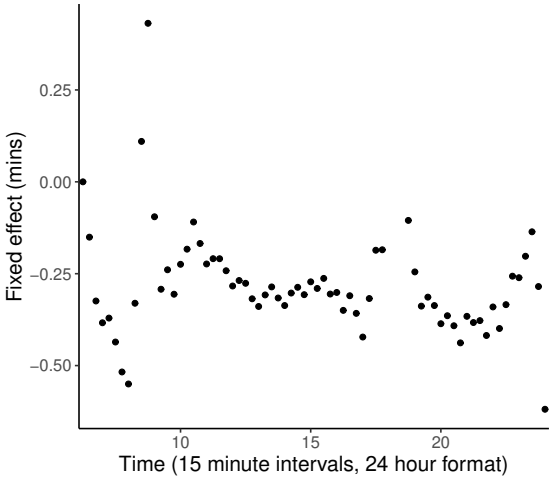


Figure 4: Fixed effects for time (mins)

Table 3: Fixed effects for passenger characteristics

Category	% Obs.	Effect (mins)
Passenger age		
Non-concession child	3E-4	1.24
Child	0.5	0.44
16-17	0.7	0.13
Adult	98.8	0.00
Passenger discount		
None	98.6	0.07
Other	0.2	0.00

Note: Obs. refers to observations. Only those categories that are significant at a minimum significance level of $\alpha = 0.01$ (99%) are reported.

The most common user demographic in the data set are passengers with adult Oyster cards and no discounts; this combined group represents 97% of all trips. Table 3 summarises the fixed effects for each significant passenger age and discount category. All four age categories are significant at a level of 99% in the model, however, only two out of the four discount types are significant at a level of 99%. The results show that children with non-concession cards (3E-4% of trips) and children with under 11 cards (0.5% of trips) have the highest age effects. These groups have relatively slower access times compared to the other groups in the data set. Those holding adult age cards and unspecified "Other" discount cards travel fastest. These groups are more likely to capture regular commuters who have a higher degree of route familiarity.

5.2. Marginal platform wait times

The following sections present the results to estimate marginal platform wait time with respect to headway beginning with a discussion of the modelling process and followed by the presentation of the results.

5.2.1. Model form

In order to understand the underlying trends in the environmental and base models of marginal wait time, regression modelling is undertaken with the derivative of access times set as the response and headway as the independent variable. Initially, non-parametric splines were fitted, however, it was found that this form resulted in over-fitting with a high degree of oscillation in the functions. To improve tractability, parametric forms were investigated and an exponential decay form was found to provide the best fit. The general form of the exponential decay function is

$$\frac{\partial E[y^{ac}(h)]}{\partial h} = \alpha + ke^{-h} + \epsilon, \tag{19}$$

where the left hand side is defined by equation (18), while k is a parametric coefficient, α is the model constant, and ϵ is the random error term.

As detailed in the literature review, many studies characterise wait times as a linear function of headway. For demonstrative purposes, the goodness of fit statistics for all eight models are presented in Appendix B where headway is modelled with a linear relationship (model form 1) and with the final exponential decay form (model form 2). As shown in the table, the exponential decay models perform better than the linear models across all criteria, indicating that this model form is more appropriate for the data.

5.2.2. Results

To begin with, it is important to understand the context of the operating conditions within the defined study area. Figure 5 shows the kernel density distribution of headways by time period. The headways between trains accommodate for the different levels in demand over the day; analysing entry frequencies by tap-in timestamps, the PM peak is the busiest time of the day, followed by the AM peak, and relatively fewer trips are undertaken in the inter-peak. Aligning with demand, headways are shortest in the PM peak with an average value of 2.40 minutes, those in AM peak are slightly longer with an average value of 2.44 minutes, and headways are longest in the inter-peak with an average of 2.94 minutes between trains.

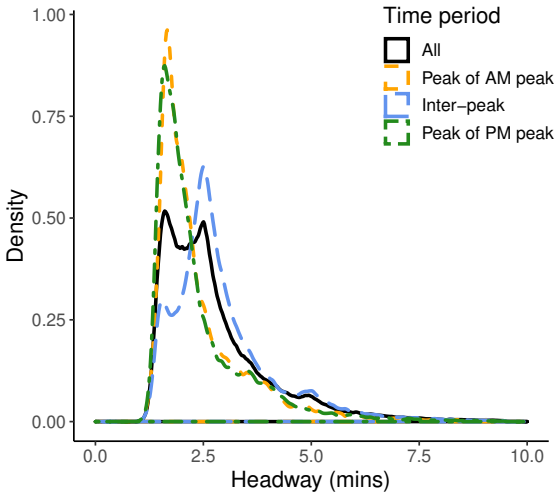


Figure 5: Distribution of headways by time period (mins)

The results refer to those for the final exponential decay form established in section 5.2.1 (model form 2 in Table B.9, Appendix B) and the regression modelling results are given in Tables 4 and 5. The environmental and base models of marginal wait times for all time periods and by time period are plotted in Figures 6 and 7, respectively, and key attributes of the functions are summarised in Table 6.

Table 4: Regression results - derivative of access times with respect to headway, base models

Model	Intercept	exp(-headway)
All	0.34***	1.16***
	(0.005)	(0.05)
AM peak of peak	0.33***	1.24***
	(0.01)	(0.06)
Inter-peak	0.33***	1.22***
	(0.01)	(0.06)
PM peak of peak	0.33***	1.20***
	(0.01)	(0.06)

Sig.: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
Standard errors in parenthesis

Table 5: Regression results - derivative of access times with respect to headway, environmental models

Model	Intercept	exp(-headway)
All	0.27***	2.45***
	(0.02)	(0.18)
AM peak of peak	0.23***	3.22***
	(0.01)	(0.16)
Inter-peak	0.26***	2.00***
	(0.02)	(0.17)
PM peak of peak	0.19***	6.15***
	(0.03)	(0.28)

Sig.: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
Standard errors in parenthesis

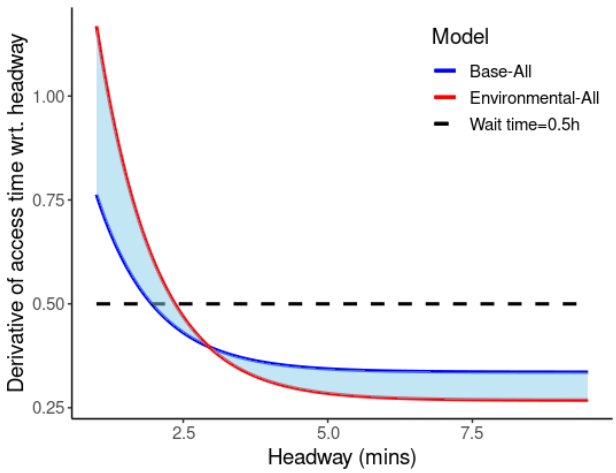


Figure 6: Derivative of access time vs headway - all time periods

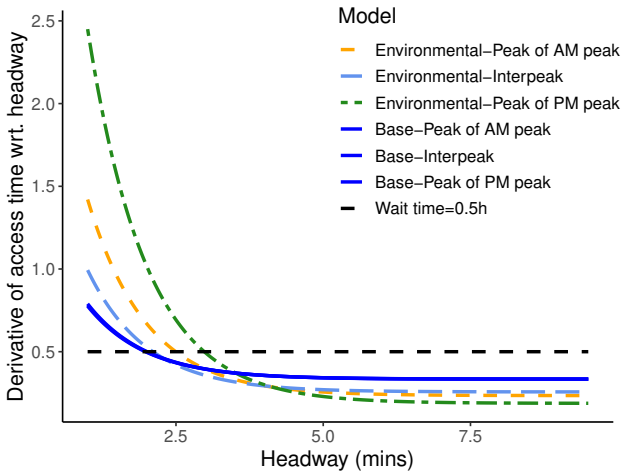


Figure 7: Derivative of access time vs headway - by time period

Table 6: Key attributes of derivative of access time functions

Model	Time period	Derivative function Max.	Derivative function Mean < 0.5	Headway at 0.5 (mins)	Proportion trips < 0.5 (%)
Base	All	0.76	0.36	1.95	72%
	Peak of AM peak	0.79	0.36	2.02	53%
	Inter-peak	0.78	0.36	2.00	82%
	Peak of PM peak	0.78	0.36	1.98	51%
Environmental	All	1.17	0.30	2.35	55%
	Peak of AM peak	1.42	0.27	2.50	31%
	Inter-peak	0.99	0.29	2.10	78%
	Peak of PM peak	2.45	0.24	2.98	21%

Comparing the results of the two sets of models, it can be seen that the effects of the different time periods present in the environmental models in Figure 7 are not observed in the base models, and across all models, the maximum values of the environmental models are greater than the corresponding maximum values of the base models. The results from the environmental and base models can therefore be considered plausible. The base models are not influenced by differing levels of congestion across the peak and off-peak time periods as the functions represent the sole influence of headway on base wait times (please refer to equation 16). The maximum values of the environmental models are higher during the AM and PM peak periods compared to the inter-peak period, and the functions cross 0.5 at longer headways during the peak periods compared to the inter-peak, implying a lower proportion of non-random arrivals during the peaks. The results for the environmental models therefore represent the differences in the environmental components of marginal wait times and walk times, including congestion and individual-specific behaviours, at different times of the day.

We now turn our attention to the form of the environmental and base models in representing marginal wait time patterns. As discussed in the literature review, the prevailing

consensus is that for high frequency services, passengers arrive randomly with a uniform random distribution. Under perfectly random conditions, the average passenger wait time (w) is equivalent to half of the headway between trains, i.e. $w = 0.5h = 0.5f^{-1}$, where h is train headway and f is the frequency of services. Under such conditions the marginal impact of headway on waiting time should remain 0.5, no matter what the initial frequency is. As shown in both the environmental and base models, when headways are less than approximately 2-3 minutes, the impact of a change in headways is much greater than half of the headway. This outcome could be a consequence of station congestion and the possibility of denied boarding during these high frequency operating periods.

In low frequency periods, the literature suggests that passenger wait times are less than half of the headway between trains i.e. $w < 0.5h$. In both the environmental and base models, when headways are longer than approximately 2-3 minutes, the models predict that the average passenger saves less time than half of the headway. This finding is in line with the literature for wait times at long headways, albeit at lower values of headway compared to the conventional rule of thumb of a 10 minute headway transition value. Across all time periods, passengers arrivals tend to become non-random at a headway between a lower bound of 1.95 minutes and an upper bound of 2.35 minutes. As a result, 55%-72% of passengers experience marginal wait times less than half of the headway. For those passengers that experience marginal wait times less than 0.5, the average marginal wait time lies between 0.30-0.36, or approximately one third of the operating headway.

Wait times below half of the headway are likely to be caused by the fact that, on average, passengers do not arrive at the station randomly. A potential explanation for this could be that when the system is operating at longer headways, the system is also less congested, and this enables passengers to move more freely through stations. The reduced levels of congestion may better enable passengers to speed up when they are aware of an approaching train, and then board the train with minimal wait time at the platform. Awareness may arise from real-time train arrival information on digital information display boards at the station prior to entry, from mobile phone applications, and/or from hearing and seeing the train approach en route to the platform.

The maximum values of the functions represent the maximum marginal wait time of passengers, and for all functions, this occurs when trains operate at the shortest headways. In this region, travel conditions are most likely to be congested with high levels of passenger demand possibly resulting in denied boarding and/or issues with train reliability with possible train bunching. For the base models, the maximum value across all time periods is 0.76 while for the environmental models the maximum values across all time periods is 1.17. The interpretation of this estimate is that if the operator manages to reduce the peak headway by one second, it leads to between 0.76-1.17 times savings in waiting time for the average passenger.

5.3. Implications for optimal supply

Section 2.3 in our literature review concluded that the marginal impact of headways on waiting time can be a crucial determinant of optimal supply, i.e. the frequency of public transport services. Given the marginal wait time estimates presented above, we are now able

to re-calibrate equation (6) and solve for the social cost minimising frequency. With the exponential specification of $\partial w/\partial h$, the optimal frequency rule of (6) becomes the following implicit function:

$$f = \sqrt{\left(\hat{\alpha} + \hat{k} \cdot e^{-60/f}\right) \frac{\gamma Q}{o'(f)}}. \tag{20}$$

Notation: $\hat{\alpha}$ and $\hat{k}$ are the parameter estimates provided in Tables 4 and 5.

Our aim in this section is to illustrate in a numerical simulation the impact that the estimated wait time model has on optimal frequency, in comparison with the usual assumption that the average wait time is half of the headway. Figure 8 plots the solution of (20) in function of the equilibrium demand level that we vary exogenously between 1 and 20 thousand passengers an hour. The remaining parameters are as follows: the value of waiting time is set to $\gamma = \$30/h$, and the unit cost of frequency provision is a constant, $o'(f) = \$200/h$. For $\hat{\alpha}$ and $\hat{k}$, we substitute the 'all day' estimates: $\hat{\alpha} = 0.27$ and $\hat{k} = 2.45$ in case of the 'environmental' model and $\hat{\alpha} = 0.34$ and $\hat{k} = 1.16$ as 'base'.

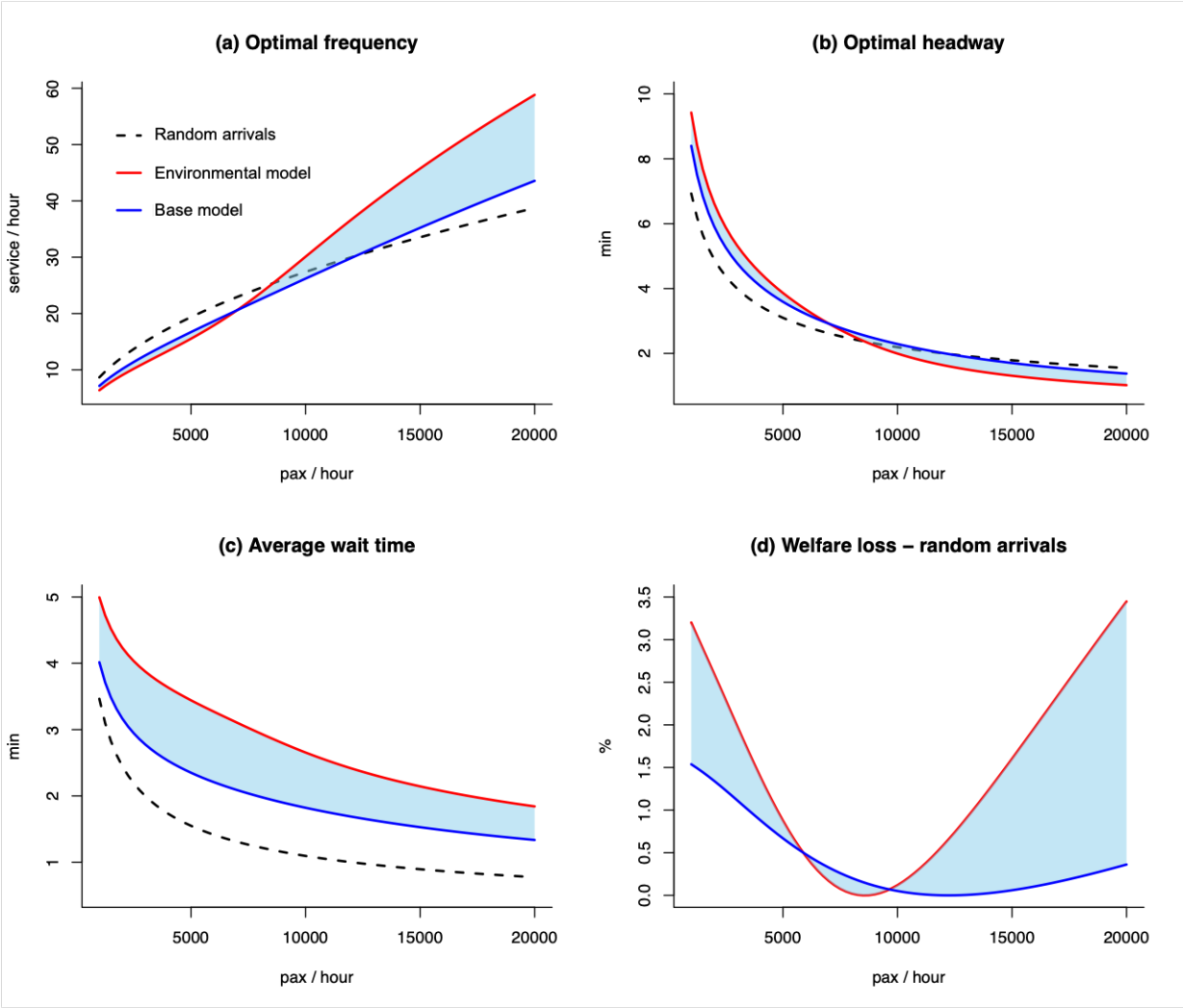


Figure 8: Properties of optimal service provision assuming (i) random arrivals, i.e. wait time set equal to half of the headway, or (ii) marginal wait time estimates of the present paper.

Panel (a) of Figure 8 reveals immediately that the replacement of the wait time assumption has striking consequences on the optimal frequency. In Section 2.3 our literature review concluded that the optimal frequency is underestimated when, in reality, $\partial w/\partial h > 0$. Figure 8 shows that this is the case when headways are short, i.e. when the operator faces relatively high demand. According to Panel (a), the degree of underestimation can be very significant: with the environmental estimates of the marginal waiting time, the optimal frequency can be around 50% higher, compared to Mohring's original square root result. However, the updated frequency curve is not always higher than the original one. At low demand and long headways, supply optimisation results in lower frequencies than before. The two curves cross each other in the range of 8,500 passengers an hour. At this equilibrium demand level, the selection of wait time model does not play an important role, as the predicted marginal

wait time is in line with the standard assumption: $\partial w/\partial h \simeq 0.5$. The base estimates do not diverge from the 'random arrivals' curve so much, but the deviation is not negligible. This curve crosses the square root function at a higher demand level, in the range of 12,000 passengers per hour.

Note that frequency is unconstrained in this exercise. In reality, technological constraints often impose an upper bound on the range of feasible frequencies. This may limit the practical relevance of non-random passenger arrivals for very-high-frequency operations.

Panels (b) and (c) are derived after straightforward transformations of the optimal frequency. The social cost minimising headway, in minutes, is simply $h(Q) = 60/f(Q)$. The average waiting time is half of the headway in Mohring's approach, while in case of this paper's model, it is the integral of the estimated marginal wait time function:

$$w(Q) = \int_0^{h(Q)} \left(\hat{\alpha} + \hat{k} \cdot e^{-h'}\right) dh', \tag{21}$$

where $h(Q)$ represents the dependency of the optimal headway on equilibrium demand. Once again, we observe in panel (b) the cut-off demand levels where the two methods produce the same headway as we would get with the random arrival assumption. At higher demand, headway should be set lower than the Mohring rule suggests, thus delivering lower waiting times for the average passenger. Interestingly, the gap in terms of average wait times with random and non-random arrivals is almost constant throughout the range of demand levels considered. The intuition is simple: headways can be allowed to increase at low demand, because passengers tend to consult the timetable, and therefore the average wait time grows less than proportionally with the headway.

The bottom-right panel of Figure 8 compares the total social cost of service provision, as defined by equation (2), with and without the assumption of random arrivals. More specifically, we compute the increase in social cost when one optimises frequency with the simplifying assumption of random arrivals, and compare it to the minimum social cost that can be achieved with the correct wait time model. The welfare loss is indeed negligible when the two approaches predict the same marginal wait time effect. However, the inefficiency induced by the oversimplified wait time assumption intensifies as we diverge from the corresponding demand level: wrong frequency setting causes welfare losses, no matter if demand is somewhat lower or significantly higher than 8,500 (environmental model estimates) or 12,500 (base model estimates) passengers per hour. With the current set of parameters and within the demand interval considered, the inaccurate assumption of random arrivals might be responsible for up to 3.5% or 1.5% increase in social costs with the environmental and base estimates, respectively. The welfare loss is typically but not necessarily higher when passenger behaviour is best described by the environmental model.

Finally, even in this simple model with inelastic demand, some conclusions can be derived about pricing and the optimal degree of subsidisation as well. Mohring's original model features very strong density economies with respect to demand, in both operating and waiting time costs. The operational cost elasticity can derived analytically directly from frequency rule (6): $\eta = \frac{\partial f}{\partial Q}\frac{Q}{f} = 0.5$, independent of the equilibrium level of demand. This implies density economies in the Mohring model, which provides justification for public subsidies

under welfare maximising pricing. By contrast, from panel (a) of Figure 8 it is immediately visible that the $f(Q)$ function is closer to a linear relationship with the updated wait time model.

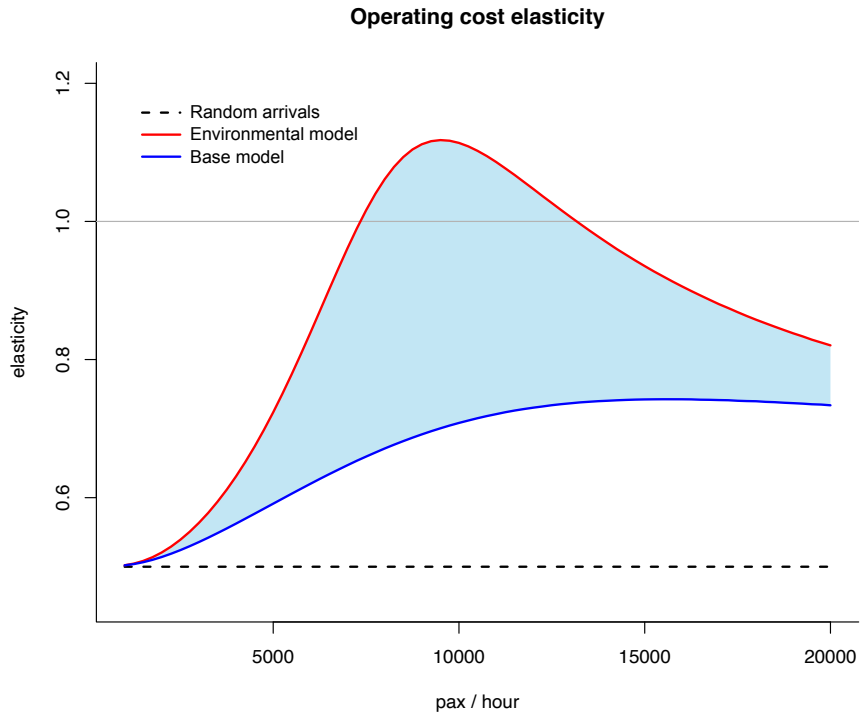


Figure 9: Elasticity of operating cost with respect to demand. As the operating cost is assumed to be linear in frequency, the values in these figure coincide with the elasticity of f w.r.t. demand.

More can be learned about the degree of density economies from the operator cost elasticities plotted separately for both models in Figure 9. The values presented here are numerical approximations of the derivative of the optimal frequency $f(Q)$. We find that the degree of density economies is milder in the entire range of demand, compared to the original model with random arrivals. Moreover, in case of the environmental estimates, there is even a demand interval in which we observe *diseconomies* of density in operating costs with respect to the final output, as $\eta > 1$ between around 7,500 and 13,000 passengers an hour. The curve of the base model does not cross the grey line of neutral scale economies, but it remains substantially higher than 0.5. This preliminary insight indicates that the prevalence of strong density economies is questionable when waiting time is modelled appropriately, and thus optimal pricing should also bring service provision closer to full self-financing. The exact value of the cost recovery ratio with a more complete micro-economic characterisation of the public transport system is an attractive subject for future research.

6. Conclusions

In this paper, the impact of headways on passenger wait times for urban metro systems is derived. The analysis is undertaken for single-line trips excluding route choice and transfer options on three lines of the London Underground metro system. First, passengers are assigned to trains to enable the total journey times recorded by the AFC data on passenger trips to be decomposed into components. After assignment, the access time component is calculated, and a semiparametric regression analysis is then performed with access time set as the response variable. Headway is specified as an explanatory variable, and additional explanatory variables are included to capture the effects of headway variance, static station and network characteristics, passenger demand, and individual passenger characteristics.

Accounting for potential confounding between the walk and wait time components of access time, we isolate the unbiased effect of headways on marginal passenger wait times via an exposure-response model framework. The results show that wait times are more impacted by changes to service frequency during high frequency operating periods, compared to low frequency operating periods. The results indicate that passenger arrival patterns transition from random to non-random as headways increase, and that marginal wait times for non-random arrivals are equivalent to approximately one third of the headway between trains. This observed behaviour could be a result of passengers consulting train arrival information prior to boarding, and if so, this would demonstrate that generalised travel time savings can be achieved through providing improved passenger information.

The empirical results are then applied in a numerical exercise of supply optimisation to show that the assumption behind the headway-wait time relationship has crucial impact on the socially optimal frequency. The evidence on non-random arrivals implies that the existing literature might underestimate the optimal peak service frequencies and overestimate those in low demand periods. Under welfare maximising service provision, the new evidence on non-random arrivals questions the prevalence of density economies in operating costs with respect to the final output (i.e. ridership), and thus also the justification for high public transport subsidies.

The exposure-response relationship of marginal wait times and headway presented here can be applied in other contexts in future work. For example, the data could be segmented at route, station, and line levels or by passenger demographics to obtain wait time patterns at a more disaggregate level. At a wider spatial scale, the analysis could be undertaken at a whole-of-network level to enable comparisons of passenger arrival patterns and wait times across different transit networks.

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Disclaimer

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References

Abkowitz, M., Tozzi, J., 1987. Research contributions to managing transit service reliability. Journal of Advanced Transportation 21, 47–65.

Aldrich, J., 1995. Correlations genuine and spurious in Pearson and Yule. Statistical Science 10 (4), 364–376.

Berggren, U., Johnsson, C., Svensson, H., Wretstand, A., 2019. Exploring waiting times in public transport through a semi-automated dedicated smartphone app survey. Travel Behaviour and Society 15, 1–14.

Bowman, L. A., Turnquist, M. A., 1981. Service frequency, schedule reliability and passenger wait times at transit stops. Transportation Research Part A: General 15, 465–471.

Brakewood, C., Barbeau, S., Watkins, K., 2014a. An experiment evaluating the impacts of real-time transit information on bus riders in Tampa, Florida. Transportation Research Part A: Policy and Practice 69, 409–422.

Brakewood, C., Rojas, F., Zegras, C., Watkins, K., Robin, J., 2014b. An analysis of commuter rail real-time information in Boston. Journal of Public Transport 18 (1), 1–20.

Csikos, D., Currie, G., 2007. The impacts of transit reliability on wait time: insights from automated fare collection system data. 86th Annual meeting of the Transportation Research Record.

Csikos, D., Currie, G., 2008. Investigating consistency in transit passenger arrivals: insights from longitudinal automated fare collection data. Transportation Research Record 2042, 12–19.

Dziekan, K., Kottenhoff, K., 2007. Dynamic at-stop real-time information displays for public transport: effect on customers. Transportation Research Part A 41, 489–501.

Fan, W., Machemehl, R. B., 2009. Do transit users just wait for buses or wait with strategies? Some numerical results that transit planners should see. Transportation Research Record 2111, 169–176.

Fosgerau, M., 2009. The marginal social cost of headway for a scheduled service. Transportation Research Part B: Methodological 43 (8-9), 813–820.

Fosgerau, M., Engelson, L., 2011. The value of travel time variance. Transportation Research Part B: Methodological 45 (1), 1–8.

Frumin, M., Zhao, J., 2012. Analysing passenger incidence behaviour in heterogeneous transit services using smart card data and schedule based assignment. Transportation Research Record 2274, 52–60.

Furth, P. G., Muller, T. H. J., 2006. Service reliability and hidden waiting time: insights from AVL data. Transportation Research Record 1955, 79–87.

Haschek, W. M., Rousseaux, C. G., Wallig, M. A., Bolon, B., Ochoa, R., Mahler, B. W. (Eds.), 2013. Haschek and Rousseaux's Handbook of Toxicologic Pathology, 3rd Edition. Academic Press, Elsevier, London, UK.

Holroyd, E. M., Scraggs, D. A., 1966. Waiting time for buses in Central London. Traffic Engineering and Control 8, 156–160.

Hörcher, D., Graham, D. J., 2018. Demand imbalances and multi-period public transport supply. Transportation Research Part B: Methodological 108, 106–126.

Ingvardson, J. B., Nielsen, O. A., Raveau, S., Nielsen, B. F., 2018. Passenger arrival and waiting time distributions dependent on train service frequency and station characteristics: a smart card data analysis. Transportation Research Part C: Emerging Technologies 90, 292–306.

Jara-Díaz, S., Gschwender, A., 2003. Towards a general microeconomic model for the operation of public transport. Transport Reviews 23 (April 2013), 453–469.

Jolliffe, J. K., Hutchinson, T. P., 1975. A behavioural explanation of the association between bus and passenger arrivals at bus stops. Transportation Science 9, 248–282.

Kittelson & Associates Inc, Parsons Brinckerhoff, KFH Group, Texas A&M Transportation Institute, Arup, 2013. TCRP Report 165: Transit Capacity and Quality of Service Manual, 3rd Edition. Transit Cooperative Research Program, Transportation Research Board, Washington, D.C.

Kutner, M. H., Nachtsheim, C. J., Neter, J., Li, W., 2005. Applied linear statistical models, 5th Edition. McGraw-Hill/Irwin, New York, N.Y.

Leurent, F., Xie, X., 2018. On individual repositioning distance along platform during train waiting. Journal of Advanced Transportation 2018, 1–18.

Luethi, M., Weidmann, U. A., Nash, A., 2007. Passenger arrival rates at public transport stations. 86th Annual meeting of the Transportation Research Board.

Mohring, H., 1972. Optimization and scale economies in urban bus transportation. The American Economic Review 62 (4), 591–604.

Newell, G. F., 1982. Applications of queueing theory, 2nd Edition. Chapman and Hall, New York, NY.

Nygaard, M. F., Torset, T., 2016. Waiting time strategy for public transport passengers. Proceedings from the annual transport conference at Aalborg University.

O'Flaherty, C. A., Mangan, D. O., 1970. Bus passenger waiting times in central areas. Traffic Engineering and Control 11 (9), 419–421.

Osana, E. E., Newell, G. F., 1972. Control strategies for an idealised public transport system. Transportation Science 6, 52–72.

Paul, E. C., 2010. Estimating train passenger load from automated data systems: application to London Underground. Master's thesis, Massachusetts Institute of Technology.

Paulley, N., Balcombe, R., Mackett, R., Titheridge, H., Preston, J., Wardman, M., Shires, J., White, P., 2006. The demand for public transport: The effects of fares, quality of service, income and car ownership. Transport policy 13 (4), 295–306.

Searle, S. R., Casella, G., McCulloch, C. E., 1992. Variance components. John Wiley & Sons, Inc., Hoboken, New Jersey.

Seddon, P. A., Day, M. P., 1974. Bus passenger waiting times in Greater Manchester. Traffic Engineering and Control 15, 442–445.

Singh, R., 2019. Decomposing journey time variance on urban rail transit systems. Ph.D. thesis, Imperial College London.

Tavassoli, A., Mesbah, M., Shobeirinejad, A., 2018. Modelling passenger waiting time using large-scale automatic fare collection data: An Australian case study. Transportation Research Part F: Traffic Psychology and Behaviour 58, 500–510.

TfL, 2013. Business Case Development Manual. Tech. Rep. V101.2013.05, Transport for London.

TfL, 2014. Geographically accurate representation of London Tube and rail lines. Tech. rep., Transport for London, accessed online at tfl.gov.uk/corporate/publications-and-reports/london-connections-map.

Turnquist, M., 1978. A model for investigating the effects of service frequency and reliability on bus passenger wait times. Transportation Research Record 663, 70–73.

Uniman, D. L., 2009. Service reliability measurement framework using smart card data: application to the London Underground. Master's thesis, Massachusetts Institute of Technology.

Wang, J., 2015. Exposure-Response Modelling: Methods and practical implementation. CRC Press, Taylor and Francis Group, Boca Raton, FL.

Wardman, M., 2004. Public transport values of time. Transport policy 11 (4), 363–377.

Wardman, M., Chintakayala, V., de Jong, G., 2016. Values of travel time in Europe: Review and meta-analysis. Transportation Research Part A: Policy and Practice 94, 93–111.

Watkins, K., Ferris, B., Borning, A., Rutherford, G., Layton, D., 2011. Where is my bus? Impact of mobile real-time information on the perceived and actual wait time of transit riders. Transportation Research Part A: Policy and Practice 45 (8), 839–848.

Wood, S. N., 2006. Generalized additive models: an introduction with R. Chapman & Hall, Boca Raton, FL.

Wood, S. N., Goude, Y., Shaw, S., 2015. Generalised additive models for large data sets. Journal of the Royal Statistical Society: Series C Applied Statistics 64 (1), 139–155.

Xie, X., Leurent, F., September 2017. Comparison of passenger walking speed distribution models in mass transit stations. In: 20th EURO Working Group on Transportation Meeting. Vol. 27 of Transportation Research Procedia. Budapest, Hungary, pp. 696–703.

Appendix A. Access time regression modelling

Table A.7: Discarded model outliers

		Raw		Filtered				
Covariate	Unit	Min.	Max.	Min.	Max.	Mean	SD	COV
Time period	15 min intervals	06:00	00:00	06:00	00:00	15:37	4.30	28%
Headway	mins	0.00	352.40	1.00	9.98	2.77	1.29	47%
COV headway	mins	0.00	3.22	0.05	1.10	0.29	0.16	56%
Norm. headway	mins	0.00	21.31	0.05	5.31	1.00	0.31	32%
Entry platform loading	number	0.01	72.00	0.30	3.00	1.40	0.69	49%
Line loading	number	0.06	4.16	0.30	3.00	1.19	0.64	54%
Passenger trips	number	1.00	793.00	1.00	35.00	8.30	9.59	116%
Access time	mins	0.00	85.87	0.10	81.47	3.49	2.04	58%

Note: SD refers to the sample standard deviation, COV refers to coefficient of variation

Table A.8: Access time models goodness-of-fit statistics

	Model 1	Model 2	Model 3	Model 4	Model 5
Continuous form	splines	splines	splines	linear	parametric/splines
Group specific form	NA	fixed	random	fixed	fixed
n	4.96E6	4.96E6	4.96E6	4.96E6	4.96E6
$R^2_{adj.}$	0.04	0.16	0.16	0.15	0.16
$D_{explained}$	0.04	0.16	0.16	0.15	0.16
AIC	2.09E7	2.03E7	2.03E7	2.03E7	2.03E7
REML	1.05E7	1.01E7	1.01E7	1.02E7	1.01E7
Headway significance	***	***	***	***	***

Sig.: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Note: n is the number of observations, $R^2_{adj.}$ is the adjusted coefficient of determination $D_{explained}$ is the proportion of deviance explained,AIC is the Akaike Information Criterion, REML is the restricted maximum likelihood score

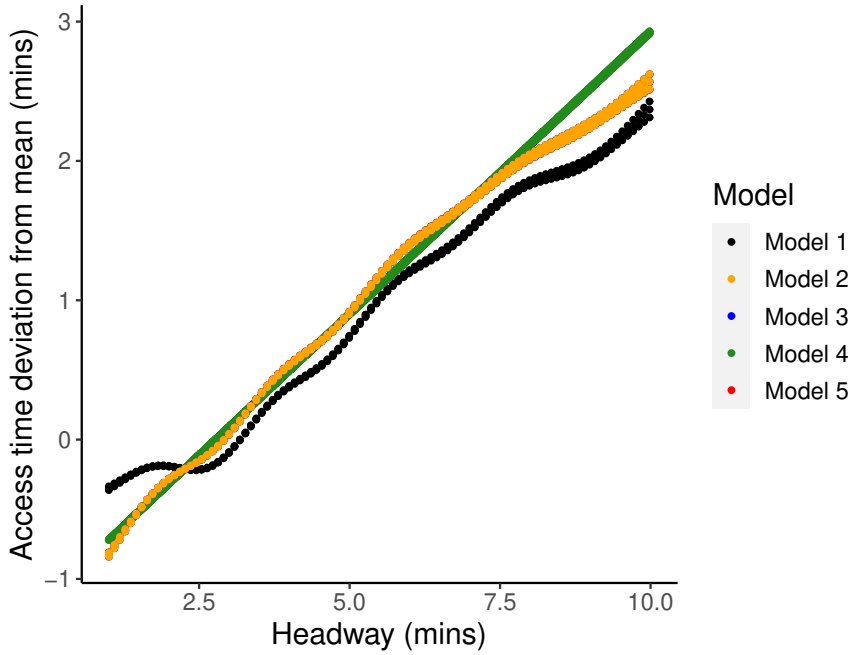


Figure A.10: Model comparison of access time vs headway relationship.
Note: model numbers refer to those in Table A.8.

Appendix B. Derivative of access time with respect to headway regression modelling

Table B.9: Derivative of access times vs headway models goodness-of-fit statistics

Model	Indicator	Model form	All	AM peak of peak	Inter-peak	PM peak of peak
Base	n	All	511	511	511	511
	$R^2_{adj.}$	1	0.43	0.41	0.41	0.42
		2	0.47	0.48	0.47	0.47
	$D_{explained}$	1	0.43	0.41	0.41	0.42
		2	0.47	0.48	0.47	0.47
	AIC	1	-892.35	-825.54	-824.79	-853.30
		2	-932.46	-891.49	-884.20	-904.50
	REML	1	-439.28	-406.01	-405.64	-419.83
		2	-462.70	-442.30	-438.67	-448.78
Environmental	n	All	511	511	511	511
	$R^2_{adj.}$	1	0.06	0.10	0.04	0.11
		2	0.27	0.43	0.22	0.49
	$D_{explained}$	1	0.06	0.10	0.04	0.11
		2	0.28	0.43	0.22	0.49
	AIC	1	395.44	426.63	325.94	1019.24
		2	264.00	190.61	222.23	739.22
	REML	1	202.09	217.62	167.48	512.77
		2	133.18	96.64	112.39	369.86

Note: n is the number of observations, $R^2_{adj.}$ is the adjusted coefficient of determination, $D_{explained}$ is the proportion of deviance explained, AIC is the Akaike Information Criterion, and REML is the restricted maximum likelihood score
Model forms as follows: 1) Linear, 2) Final form - Exponential decay.